SAGIP: Solar-Powered Estero Guardian with On-Chip AI

Quezon City Science High School

Chosen Theme:
- Environmental Sustainability through Technology

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Our device is a low-cost, solar-powered unit that lifts floating plastic out of a creek and counts each item with a tiny on-chip AI model. Barangay officials get a weekly offline report of what was removed, plus a secondary water-level alert for blockages. Built for about PHP 6,000 from local parts, it adds anonymized records to an open dataset that other barangays can copy.

1. What problem are we trying to solve?

Philippine rivers rank among the largest sources of ocean plastic (Jambeck et al., 2015, Lebreton et al., 2017, Meijer et al., 2021), and small urban esteros feed them. Cleanups weigh bulk waste and record nothing about it. Brand audits are periodic and mostly land-based (Break Free From Plastic, 2019), and RA 11898 needs proof of which packaging reaches waterways. The gap is a continuous, low-cost record taken at the creek itself.

2. What do we want to achieve?

Goal: make creek plastic measurable evidence instead of invisible waste, with cleaner creeks and community-owned data.

Objectives: (1) In an 8-week pilot from December 2026, lift at least [TARGET] kg of floating debris and capture at least [X]% of what reaches the boom. (2) By November 20, reach 85% precision and recall on unseen photos, with weekly counts within 15% of a hand audit. (3) By March 2027, publish an open dataset and build guide and help [2] more barangays or schools build a PHP 6,000 unit.

3. Who will benefit from this project?

Primary: the [N] households of [barangay and purok] along [estero] and the barangay sanitation officers. **Secondary:** the city solid-waste office and DENR, [2] more barangays or schools, and downstream fisherfolk.

4. How did we come up with this proposal?

Select. We scored five community problems on significance, feasibility, STEM depth, replicability and originality. Creek plastic won. [Edit to match your process.]

What exists. Cameras count floating plastic but remove nothing (van Lieshout et al., 2020). Robotic boats collect it but need radio and an operator (HydroSentinel, 2026). Trash traps and interceptors remove it in bulk with little item-level data (Manila Bulletin, 2026, Schmaltz et al., 2020). Brand audits name companies but are manual (Break Free From Plastic, 2019).

Gap we test. No system we found lifts, counts and shares open data at barangay cost. Can a PHP 6,000 on-chip counter match a hand audit within 15%, and does continuous counting show rain-driven pulses of plastic (Cordova & Nurhati, 2019) that weekly audits miss?

Sources: Break Free From Plastic and GAIA 2019, Cordova & Nurhati 2019, David et al. 2021, HydroSentinel 2026, Jacob et al. 2018, Jambeck et al. 2015, Lebreton et al. 2017, Manila Bulletin 2026, Meijer et al. 2021, RA 11898, Schmaltz et al. 2020, van Lieshout et al. 2020.

5. How will this tech solution work?

Summary. A floating boom guides plastic to a bar screen, a camera-triggered arm lifts it into a bin, and an ESP32-S3 camera counts each item on the chip with an INT8 model (Jacob et al., 2018, David et al., 2021). An offline Wi-Fi page gives the barangay a weekly report and an open dataset. An ultrasonic sensor warns of rising water as a secondary function. A shear pin, current and Hall sensing and a three-retry lockout protect against jams.

STEM. Physics sets buoyancy, torque and the solar budget. Mathematics defines precision, recall and count error. Computer science supplies the quantized model. Engineering joins them in a weatherproof floating structure.

Key message. Proof of what plastic reaches a creek can be continuous, cheap and community-owned, and one chip delivers cleanup and evidence together.



6. How do we implement this?

A pilot unit costs about PHP 6,000, built in three stages that are each bought only after the previous test passes.

Resources: ESP32-S3 camera board 1,300, microSD 200, box 300, solar panel 600, charger and cells 700, boom and post 800, bin and gauge 150, wiring 250, level sensor 350, motor and driver 700, sensing 150, ramp and arm 150, contingency 350. Stages: bench 1,800, passive trap 2,500, arm and sensor 1,350. Timeline: model by Nov 20, results Dec 7, pilot Dec to Feb, guide by March.

Stakeholders: barangay captain and sanitation officers (site approval, weekly emptying, one-page agreement), city solid-waste office and DENR (use the data), adviser and school head (supervision, safety), and residents (consent).

Risks & Mitigation: Hyacinth mats: deflector and a vegetation class that stops the arm. Jams: shear pin, current and Hall sensing, three retries. Oil and silt: motor mounted high, sealed bearings. Model errors: test on unseen places and keep the hand count. Arm failure: boom, bin and camera still work. Safety and privacy: no one enters the water and the camera sees water only.

7. What kind of impact do we hope to achieve and what is the long term vision for this proposal?

Success means plastic lifted from a real creek and a trustworthy weekly record of what it was.

Measurable Outcomes: (1) At least [TARGET] kg lifted in 8 weeks, with a capture rate of at least [X]% against a baseline week. (2) Weekly counts within 15% of the hand audit, and 85% precision and recall on unseen places. (3) At least [2] more sites by March 2027 at about PHP 6,000 each, sharing one open data format.

Long-Term Vision: a student-built, barangay-owned network of units along connected creeks, feeding one open dataset that shows where plastic enters, which products dominate and whether EPR commitments change what the water carries. Planned upgrades are a belt conveyor and a second camera angle.

Key Takeaway: the Philippines already knows who makes much of its plastic. SAGIP makes the proof continuous, cheap and community-owned, and lifts some of the plastic out along the way.